

Hardware Assessment Report

Bostontech India

Scan Date: June 11, 2026 at 08:55 AM



Estimated Lifespan: 2-3 years

System Information

Hostname: DESKTOP-TRAO6V6

OS: Microsoft Windows 11 Pro 11

CPU: 12th Gen Intel(R) Core(TM) i7-12700

RAM: 31.68 GB

Overall System Health: 71/100 (Grade B)

Estimated Remaining Lifespan: 2-3 years

No Critical Issues Detected

Component Health Summary

Component	Score	Grade	Status
12th Gen Intel(R) Core(TM) i7-12700	80	A	No action needed
31.68 GB (2 modules)	90	A	No action needed
TOSHIBA DT01ACA100	73	B	No action needed
IM2P33F8-256GD4	74	B	No action needed
Intel(R) UHD Graphics 770	64	B	No action needed
HP 8950	77	B	No action needed

CPU — 12th Gen Intel(R) Core(TM) i7-12700

Grade A (80/100)

Part/Serial: BFEBFBFF00090672 Status: ■ OK

→ Reapply thermal paste to reduce operating temperatures

→ Clean CPU heatsink and fan to prevent thermal throttling

RAM — 31.68 GB (2 modules)

Grade A (90/100)

Part/Serial: M323R2GA3BB0-CQKOL Status: ■ OK

→ Run MemTest86 to verify memory stability and rule out bad sectors

→ Ensure RAM is configured in dual-channel mode for optimal bandwidth

Storage — TOSHIBA DT01ACA100

Grade B (73/100)

Part/Serial: 72E6M2VMS Status: ■ OK

→ Defragment the drive regularly to maintain read/write performance

→ Monitor S.M.A.R.T. values weekly to detect early signs of failure

Storage — IM2P33F8-256GD4

Grade B (74/100)

Part/Serial: 0000_0000_0000_0000_0000_0000_0000_... Status: ■ OK

→ Enable TRIM support in the operating system for optimal endurance

→ Avoid filling the SSD beyond 80% capacity to extend lifespan

GPU — Intel(R) UHD Graphics 770

Grade B (64/100)

Part/Serial: Intel(R) UHD Graphics Family Status: ■ OK

→ Integrated GPU is soldered to the CPU/motherboard — cannot be upgraded separately

→ To improve GPU performance, a motherboard + CPU replacement is required

Motherboard — HP 8950

Grade B (77/100)

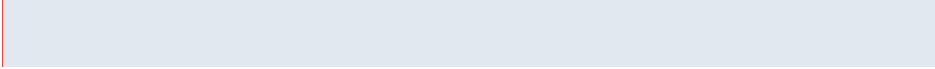
Part/Serial: PMNHD0BCYGK166 Status: ■ OK

→ Update BIOS firmware to the latest version for compatibility fixes

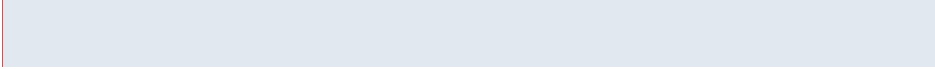
→ Visually inspect capacitors for bulging or leaking (sign of failure)

Overall Benchmark Score: 0/100

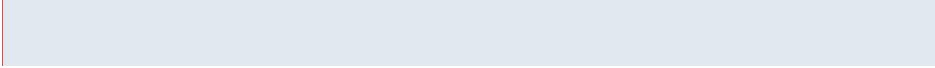
CPU Compute0/100



Disk I/O0/100



Memory Bandwidth0/100



Category	Status	Bottlenecks
Web Development	Fully Supported	None
Desktop Development	Fully Supported	None
Mobile Development	Fully Supported	None
Databases	Fully Supported	None
Virtualization	Minimum Requirements Only	Recommended 32GB RAM (has 31.68GB)
AI & Data Science	Not Supported	Recommended 32GB RAM (has 31.68GB); Needs 4096MB VRAM (ha...
CAD & Engineering	Minimum Requirements Only	Recommended 32GB RAM (has 31.68GB); Recommended 4096MB VR...
Game Development	Not Supported	Recommended 32GB RAM (has 31.68GB); Needs 4096MB VRAM (ha...
Video Editing	Minimum Requirements Only	Recommended 32GB RAM (has 31.68GB); Recommended 6144MB VR...
Basic Office & Browsing	Fully Supported	None

Total Recommendations: 4

- [HIGH] System
Recommended 32GB RAM (has 31.68GB)
- [HIGH] System
Needs 4096MB VRAM (has 2048.0MB)

Total Estimated Upgrade Cost

Rs. 25,500 INR

(~\$280 USD)

Prices as of 2026-04-07 | Sources: Amazon.in, MDComputers, PrimeABGB

Priority	Component	Current	Recommended Upgrade	■ INR	\$ USD
✓ GOOD	Storage	931.51GB HDD (TOSHIBA	1TB SATA SSD	Rs. 7,500	\$80
OK	CPU	12th Gen Intel(R) Core	Intel i5 (12th/13th Gen)	Rs. 18,000	\$200
			TOTAL	Rs. 25,500	\$280

Upgrade Justification

- Storage: Replace 931.51GB HDD with 1TB SATA SSD (3-5x speed boost)
Source: Amazon.in (<https://www.amazon.in/s?k=1tb+ssd>)
- CPU: Upgrade from 12-core/2100.0MHz to modern mid-range CPU
Source: MDComputers (<https://mdcomputers.in/processors?brand=intel&tier=i5>)

* Prices are approximate and may vary. Consult your vendor for exact quotes. Currency rate: 1 USD = ■92.0